

# Multi-Robot Coordination in Dynamic Environments Shared with Humans

Zeynab Talebpour and Alcherio Martinoli

**Abstract**—This work addresses multi-robot coordination in social human-populated environments using a market-based framework for solving the Multi-Robot Task Allocation (MRTA) problem. Humans are considered in the proposed coordination mechanism by means of accounting for social costs in bid evaluations and requesting collaboration in socially blocking situations. Initially, the effect of a realistic environment with varying number of static/moving humans on the behavior and performance of our method is studied through an extensive suite of experiments in a high-fidelity simulator. Results show that the total traveled distance and time are increased when humans are present in the environments. Localization noise is also increased particularly in the case of static people. In the second series of experiments, a number of problematic cases resulting in longer modified paths, blocked passages, and long waits have been investigated. A comparative study targeting human-agnostic navigation and planning, human-aware navigation and human-agnostic planning, and human-aware navigation and planning has been conducted. Both simulated and real robot experiments confirm the effectiveness of accounting for humans at both team and individual levels. This leads to respecting social constraints as well as achieving a better performance based on MRTA metrics.

## I. INTRODUCTION

Robots are progressively becoming a part of our environment. Therein, they either directly interact with people or have to operate in spaces shared with humans. It is of utmost importance that people co-existing with robots feel safe and comfortable around them [1], and perceive their actions as legible and predictable [2]. Thus, for long-term sustainable deployment in social environments, it is essential that robots gain the acceptance of people.

Navigation is one of the main required functionalities for enabling robots to be actively used in real social environments. There, the quality of the movement of robots strongly influences how their intelligence is perceived [3]. “The intersection between research on human-robot interaction (HRI) and robot motion planning” [1], known as human-aware navigation, focuses on finding solutions for socially acceptable robot navigation [4], [5]. However, the research in this area is mostly considering single robot applications and the problem of cooperative human-aware navigation for Multi-Robot Systems (MRS) -an interesting problem for both multi-robot and human-aware navigation research- is largely unexplored.

The main goal of this paper is to provide insights into one particular class of MRS coordination mechanisms commonly known as Multi-Robot Task Allocation (MRTA) [6], [7] in social environments. In such environments the number of robots are often limited and the number of tasks are usually moderate (e.g., assistant robots operating in homes or hospitals). The main difficulty for MRTA in such highly dynamic and noisy environments is that plans are likely to

change or to be rendered invalid, particularly if the robots are planning for long periods of time. Additionally, the robots are required to perform in a socially acceptable manner in terms of navigation and interaction with people and other team members. This adds additional constraints to the planning problem.

Robots can benefit from coordination when faced with social constraints. As an example, when a human chooses to block a passage that a robot needs to traverse for reaching a desired destination to perform a task, what is the best strategy for the team? The human might want to start an interaction with the robot. In this case, the robot will no longer be available to perform the team plan. It could also be that the human has no intention for interaction but simply needs to use that space, for instance because he or she is having a conversation in front of a door. By assigning the already taken task to another robot through coordination, the team level plan can change on demand to adapt to this situation. Accounting for social factors in task assignment can lead to better team plans in terms of social acceptability and even performance metrics such as time, if team level planning considers the social costs and handles the unforeseen changes in the environment by means of coordination.

To test these hypotheses, studies involving both MRTA and social performance metrics are required in environments with different levels of complexity in terms of noise and dynamics of humans. To the best of our knowledge, human-aware robot coordination for team level planning in MRTA has not been investigated in the literature. The contributions of this work include, providing insights on human-agnostic MRTA in social environments of varying complexity, proposing a social MRTA method that accounts for humans in both individual and team level plans, and conducting a comparative study in simulation and with real robots for MRTA instances with: 1) no humans present, 2) humans considered only as unmapped obstacles, 3) humans considered as social entities by individual robots at navigation level only, and 4) humans considered both at task planning and individual navigation levels.

## II. RELATED WORK

MRTA algorithms vary in design and application but their common objective is to find a mapping between robots in a team and a set of “tasks” that must be accomplished in order for the team “goal” to be completed. Among multiple approaches proposed for MRTA, we are mainly interested in distributed approaches that can be executed by a team of robots without the explicit need for an external centralized entity with perfect knowledge of the environment [8].

Market-based multi-robot coordination [9]–[12] is an example of such an approach. In these systems, robots act as agents trying to maximize their individual profits. Every time a task is auctioned, robots must pay a price to obtain it. Once the task is completed, a payment is done to the robot which won the auction. The underlying assumption is that

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with every robot trying to maximize its individual profit, the overall team coordination and efficiency will be improved.

Single or multiple tasks can be auctioned in the context of market-based coordination. Sequential Greedy Algorithm (SGA) and Consensus-Based Auction Algorithm (CBAA) proposed in [13] are examples of single task auction algorithms. SGA is a centralized method that assigns the task with the minimum cost to each available robot, thus, adopting a greedy approach. Similar solutions to SGA can be achieved in a decentralized manner by CBAA through auction and consensus mechanisms. Auctioning multiple tasks is enabled by the Consensus-Based Bundle Algorithm (CBBA) proposed in [13] and by the Hoplites framework [14] by means of coordinating “plans” encapsulating multiple tasks. The Hoplites framework consists of two concurrent coordination mechanisms: passive and active. A passive coordination quickly produces locally-developed solutions while active coordination produces complex team solutions via negotiation among teammates. Active coordination makes the distinction between this approach and CBBA. The active coordination scheme allows for more flexibility in Hoplites compared to CBBA, since robots can modify their plans after having been allocated a task or even in action, upon request of other team members. This request is formed locally by the robots and targets only a subset of team members that are locally identified for collaboration.

In our previous work, a study of MRTA in dynamic and noisy environments [15] for spatial task allocation confirmed the effectiveness of our Hoplites-based proposed method in comparison to SGA and CBAA as a first step towards deployment of MRTA methods in social environments. Due to the highly dynamic and noisy nature of such environments, planning for long horizons will not be effective and could incur unnecessary computational costs without contributing much to a better plan compared to shorter planning horizons. Therefore, short planning horizons in terms of plan lengths were chosen and compared to single auction MRTA methods. Results confirm that the proposed method can achieve assignments that require less traveled distance and therefore take less time, as well as creating a well-balanced load for the team, when tasks are uniformly distributed in the environment.

MRTA in social environments should ensure social acceptance as well as achieving good performances in terms of global team objectives such as the total traveled distance, mission time, etc. Respecting personal spaces, O-spaces and P-spaces [16] are the common social behaviors considered in the literature of human-aware navigation based on the concept of “proxemics” [17]. In this work, proxemics-based social costs encoded as costmaps similar to [18] along with a Fast Marching Method (FMM)-based path planner constitute the human-aware navigation components of the robots. FMM has been proven to be successful in real domestic spaces with high complexity [19]. There have previously been a number of research papers addressing social path planning using FMM [18], [20].

### III. PROBLEM DESCRIPTION AND APPROACH

We propose a system that builds upon two main components of Hoplites-based multi-robot coordination [15] and human-aware navigation [21], both already verified in our previous work. In this section, we will revisit these components and highlight the extensions and improvements

motivated by social human-populated environments leading to our proposed method.

#### A. Problem Description

Consider a team of robots and a set of tasks, where each task is a location that a robot has to visit. The team of robots should decide how to efficiently subdivide these tasks using a local criterion in order to optimize a global performance. This global performance can be a function of time, distance traveled, etc. Tasks can be identified locally by the robots through on-board perception or can be broadcasted to all robots by an external source. Many applications such as patrolling, attending service requests, etc., can benefit from this functionality.

For succeeding in a social environment, robots are required to ensure that social constraints, namely personal and interaction spaces of humans, are not violated.

#### B. Hoplites-based Multi-Robot Coordination

In [15] we showed that using a revenue function  $R$ , and a cost function  $C$ , the robots, which are self-interested agents pursuing individual profit, can evaluate each available task and decide whether to take it or sell/buy it to/from another robot in a distributed manner by means of the built-in coordination mechanism. The strategy employed in [15] consists of two main concurrent types of coordination: *passive coordination* and *active coordination*.

In **passive coordination**, each robot chooses its most profitable plan and broadcasts it to other teammates without any attempt to modify their plans. This information is then used by other robots to reevaluate the expected profitability of their current plans, update and broadcast the changes. However, sometimes a robot's best plan can only be marginally profitable while a genuine team plan could result in a higher profit. This suggests that modifying the plans of the robot's teammates would be an interesting option to pursue. Concretely, this implies that the requesting robot asks its collaborators for compensation price quotes and persuades them to engage in a cooperative action. This process is ruled by a market-based approach and constitutes the **active coordination**. The decision of switching to the active coordination mode is based on the evaluation of a **local balance function**.

Details of this algorithm can be found in Alg. 2 of Section III-D, where we present the extensions and improvements made to the above approach to meet the requirements of a social environment. The local balance function is strongly problem-dependent and can contribute to reaching the globally optimal solution at the team level if chosen correctly. Additionally, a problem-dependent **global balance function** is also required for team-level evaluations. Details of the local and global balance functions will be described in Section IV.

#### C. Human-aware Navigation

The personal space around a human can be modeled as a mixture of two Gaussian functions, one in the front and another in the rear of the area surrounding the person [18]. The orientation and heading of the human will cause a corresponding rotation of these functions in such a way that the person is always in the center and the absolute orientation of the human matches that of the Gaussian functions. A

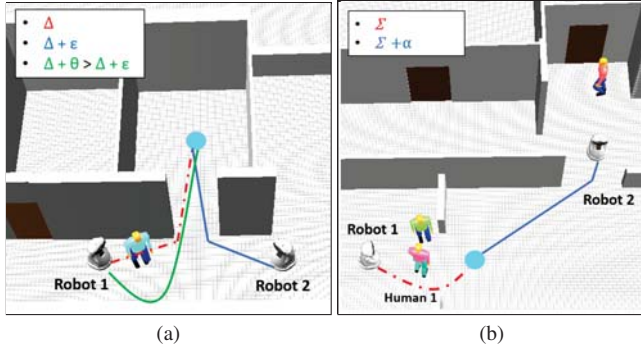


Fig. 1: Two cases where accounting for the presence of humans lead to a modified team plan. a) Accounting for the added distance caused by human-aware navigation in the local balance function. b) Social time-out occurring as the result of humans forming a blocking passage. The cyan circle represents the task.

Gaussian function  $\phi$ , centered on  $p$  with covariance matrix  $S$ , is defined as follows:

$$\phi(q) = \exp\left\{-\frac{1}{2}(q-p)^T S^{-1}(q-p)\right\} \quad (1)$$

$q$  indicates the position of a point and  $S$  is:

$$S = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix} \quad (2)$$

The center of the costmap indicates the position of the person and  $\sigma_x$  and  $\sigma_y$  are used to modulate the shape of the Gaussian and are traditionally chosen in a way to respect the personal space of a person as indicated by the proxemics principle. Various factors can influence the size of this area [21], however  $\sigma_x$  and  $\sigma_y$  are commonly considered to be constant. Getting closer to a person, will cause an increase in the value of this function, and the social cost associated to that position will increase as a consequence.

As shown in our previous work [21], augmenting the path planner with social cost information modeled as social costmaps can lead to successful human-aware navigation for individual robots. Similarly, in this work updates about human positions are regularly sent to the FMM path planner in the form of Gaussian cost functions and using the concept of layered costmaps introduced in [22], this information is then used for performing the replanning.

#### D. Social Multi-Robot Coordination

In the context of social environments, the presence of people can largely modify the team plans if social constraints are to be accounted for. If the environment is changing in terms of human positions and interactions, how should a robot estimate the plan cost? What should be the strategy if a plan is rendered invalid due to the changes in the environment? How often or when should the robots replan and recompute the costs? These are interesting and challenging questions that need to be considered in MRTA for human-populated environments.

In this work, we introduce two extensions to the previous market-based coordination method.

1) *Socially-aware Balance Functions:* Since a distributed auction process requiring bid computations governs the market, accounting for social costs entails improving and extending the estimation of the balance function to include a

social term. Hence, social factors should also be aggregated into the estimated value for bidding on the tasks. Designing the balance functions and aggregating the social terms are problem-specific and may vary across different environments.

Compared to a human-agnostic bidding, the bid estimates containing social terms can result in modified team plans. As an example, in Fig.1a, given a local balance function that scores tasks inversely proportional to their distance to the robot, Robot 1 will take the task since its path length ( $\Delta$ ) to the task is inferior to that of the other robot ( $\Delta + \epsilon$ ), with  $\epsilon > 0$ . However, by taking into account the added distance  $\theta$  that Robot 1 has to travel to avoid intruding the personal space of the human, Robot 2 will take the task since  $\Delta + \theta > \Delta + \epsilon$ , with  $\theta > \epsilon > 0$ .

2) *Social Time-outs:* If humans start interacting with a robot that has been assigned a task, assuming that attending to immediate requests of humans has a higher priority, the robot should stop. In a human-agnostic planning approach other team members assume that the task is taken and carry on with their remaining tasks. If a time-out is foreseen for unfinished tasks, after waiting for that period of time the task would be available on the market again and otherwise other robots will assume it will eventually be reached by the responsible robot. A similar situation can happen if the cost of a task changes due to the movement of the humans. As illustrated in Fig.1-b. When robots are scoring the task while considering the social costs, Robot 1 will be the best candidate given the smaller distance to the task location since  $\Sigma < \Sigma + \alpha$ ,  $\alpha > 0$ .

However, if human 1 walks back while maintaining the interaction, the path of Robot 1 can be blocked by the larger interaction space formed by the two humans (O-space). This way, despite Robot 1 having enough free space to pass, it will have to cross the O-space and thus, incur a larger social cost or it might have to wait for a long time trying to find a new path as people move. Given the formulation chosen for the local balance function and the weight of the social term, Robot 2 can be in such situation a better candidate for taking the task.

We have introduced a priority planning caused by social time-outs in our proposed coordination algorithm. In this work (see Alg. 1), social time-outs are triggered when immediate human interactions start with an active robot or in case an active robot is not advancing in its planned path in the vicinity of humans for a predefined period of time. If a robot detects a social time-out it broadcasts collaboration requests through active coordination.

Without going into the details of active and passive coordination reported previously in [15], Alg. 2 describes our proposed market-based coordination method. While unattended tasks are available, each robot computes a plan which is a sequence of tasks by means of passive coordination. This plan is stored if the robot is currently busy performing another task. If the plan computed through passive coordination results in a low revenue or if social time-outs are detected, active coordination is triggered and other robots are asked to cooperate. It is assumed that a group of robots will only accept to cooperate if the requesting robot is able to pay all of them compensation prices. Turn taking and priority planning govern the sequence in which robots compute and broadcast the unattended plans. Additionally, when cooperation requests are received and granted, the priority for all cooperating

robots will be to replan to accommodate the change in the team plan. Accounting for social costs in bid computation and requesting collaboration upon social time-out detection are the main extensions to our previous work.

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**Algorithm 1** Social time-out detection for robot  $r_j$

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1: procedure SOCIALTIMEOUT( $Tasks, r_j$ )
2:    $TimeOut \leftarrow False$ 
3:    $\triangleright$  While the robot is assigned a task and is moving towards it
4:   while ISACTIVE( $r_j$ ) do
5:      $\triangleright$  If at least one human is present close to the robot
6:     if INHUMANVICINITY( $r_j$ ) then
7:        $\triangleright$  If at least one human has started interaction with the robot
8:       if HUMANINTERACTIONREQUEST( $r_j$ ) then
9:          $TimeOut \leftarrow True$ 
10:     $\triangleright$  If the robot has not moved enough during a time window
11:    if NAVIGATIONPROGRESSED( $Tasks, r_j$ )  $\neq True$  then
12:      if HUMANINTERACTIONREQUEST( $r_j$ )  $\neq True$  then
13:         $TimeOut \leftarrow True$ 
return  $TimeOut$ 

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**Algorithm 2** Market-based coordination for robot  $r_j$

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1:  $\triangleright$  This is the procedure running on each robot
2: procedure MARKETBASEDCOORDINATION
3:    $currentPlan \leftarrow \emptyset$ 
4:    $storedPlan \leftarrow \emptyset$ 
5:    $coopAccepted \leftarrow 0$ 
6:    $\triangleright$  While there is a task to be assigned
7:   while NOTEMPTY(GETTASKS()) do
8:     if MYTURN( $r_j$ ) then
9:        $\triangleright$  Check if the stored plan is valid
10:      if ISVALID( $storedPlan$ ) then
11:         $currentPlan \leftarrow storedPlan$ 
12:      else
13:         $newPlan \leftarrow \emptyset$ 
14:         $\triangleright$  Get all tasks regardless of their availability
15:         $Tasks \leftarrow GETTASKS()$ 
16:         $newPlan \leftarrow PASSIVECOORDINATION(Tasks, r_j)$ 
17:         $\triangleright$  Compute the revenue
18:         $R \leftarrow GETREVENUE(newPlan)$ 
19:        if  $R \leq Thr \mid$  SOCIALTIMEOUT( $Tasks, r_j$ ) then
20:           $activePlan, R_c, Plans \leftarrow ACTIVECOORDINATION(Tasks, r_j)$ 
21:           $coopAccepted \leftarrow ASKFORCOOP(R_c, Plans)$ 
22:          if ( $coopAccepted$ ) then
23:             $newPlan \leftarrow activePlan$ 
24:           $currentPlan \leftarrow newPlan$ 
25:        else
26:           $storedPlan \leftarrow PASSIVECOORDINATION(Tasks, r_j)$ 
27:          if RECEIVEDCOOPERATIONPROPOSAL() then
28:            EVALUATEPROPOSAL()
29:            BROADCASTANSWER()

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#### IV. METRICS AND NAVIGATION

Evaluation metrics and an overview of our navigation system are detailed in this section.

##### A. Metrics

Robots should find the most appropriate assignment for optimizing the global balance function based on a local balance function. The local balance function of robot  $r_j$ , for each task  $t_i$  belonging to a plan  $P$  is defined in the following. This function is inversely proportional to the length of the path planned by the FMM to the desired task. We note that for social task planning, the path planned by the FMM already takes into account the social costmaps representing the human-centric Gaussian cost functions.

$$B_{j,P} = \sum_{t_i \in P} (R_{t_i} - \text{human\_aware\_distance}(\text{pos}_{t_{i-1}}, \text{pos}_{t_i})) \quad (3)$$

$R_{t_i}$  is the revenue of task  $t_i$ ,  $\text{pos}_{t_i}$  is the position of  $t_i$  and  $\text{pos}_{t_{i-1}}$  is the position of the task that appears before  $t_i$  in  $P$ . This function includes a revenue  $R_{t_i}$  that is decreasing with time as shown below.

$$R_{t_i}(t) = \max(0, R_{max}(1 - \frac{t - t_{a,i}}{T})) \quad (4)$$

where  $t$  is the time in which  $t_i$  is reached,  $R_{max}$  is the maximum revenue for the task,  $t_{a,i}$  is the allocation time of  $t_i$  and  $T$  is the time after which the positive revenue becomes zero. This utility function is added to reinforce reaching the tasks as early as possible. In real noisy social environments, only an estimate of the time to reach a task can be computed. Additionally, due to the presence of people, because of the random interactions they form among themselves and with robots, estimations of the task times can be far from reality for some of the robots and acceptable for those that are not faced with humans. In this paper, we decided to only include the distance term in the local balance function.

As for the global balance function concerning MRTA, the total traveled distance ( $M_1$ ) and the mission time ( $M_2$ ) are reported for all experiments. For evaluating the performance of the MRS in terms of social-awareness, the maximum accumulated social cost ( $M_3$ ), the maximum time steps spent in areas associated with social costs ( $M_4$ ) and the minimum distance to any human throughout the experiment ( $M_5$ ) are reported among all robots.

##### B. Navigation

The navigation system is that of the MONARCH project [23], detailed in [19]. As input, it uses the pose estimates provided by a standard Adaptive Monte Carlo Localization (AMCL) self-localization system, given odometry, laser range finder readings, and a static map. The navigation system is based on FMM for motion planning, together with a Dynamic Window Approach (DWA) algorithm for guidance and obstacle avoidance. FMM and DWA run asynchronously. FMM is activated when a new goal position is given, and DWA is running in a closed loop with a fixed rate of 20 Hz in our experiments.

#### V. EXPERIMENTS

In this section we will describe the robots used for our experiments, the simulation tool and the experimental setup.

##### A. Robot

The robot we used in this work is called the MBot [24] (see Fig. 4) and has been developed within the FP7 European project MONARCH<sup>1</sup>. It is an omni-directional drive robot with an approximately round footprint of 0.65 m in diameter and a height of 0.98 m. It is endowed with two laser range finders, on both the front and the back for providing 360° coverage.

##### B. Simulations

The use of high-fidelity simulators such as Webots [25] is fundamental, especially when considering multi-robot systems. When evaluating MRS in human-populated environments, conducting real experiments with varying features and complexity levels becomes exceeding difficult. We have developed models of the environments (see Fig. 2 and Fig. 3) for our experiments that enable simulations with similar environmental richness to the real world experiments, as shown in [15].

<sup>1</sup><http://monarch-fp7.eu/>

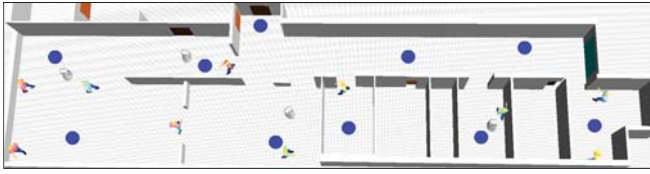


Fig. 2: Snapshot of a sample test for ten moving humans in Webots. There are ten tasks depicted by blue circles representing the locations that should be visited by the robots.

### C. Experimental Setup

Three case studies of varying complexity have been addressed in this work. The following sections describe them in detail.

1) *Human-agnostic MRTA in Social Environments*: To get a better understanding of how the existing MRTA methods perform in human-populated environments without considering any additional social elements, a series of experiments have been conducted. We have tested five different scenarios in simulation with the following configuration in terms of presence of people, *Scenario 1A*: no humans, *Scenario 1B*: five static humans, *Scenario 1C*: five moving humans, *Scenario 1D*: ten static humans, and *Scenario 1E*: ten moving humans.

Static humans are randomly placed on positions that the robots would have to traverse. Dynamic humans are randomly moving in all parts of the environment while avoiding collisions with other agents. Fig. 2 shows the faithful simulated model of the oncological ward of the IPOL<sup>2</sup> hospital in Lisbon used in our simulations as a representative complex indoor environment. Robots are relying on their self-localization for computing the local balance functions and the evaluation metrics have been obtained from ground truth values provided by the simulation. Initial positions of robots and humans for all experiment sets are the same. The reason for this choice is firstly, to compare the same MRTA problem instance across all scenarios and secondly, to ensure that humans are located at positions that are likely to create social costs for the robots.

In this case study, a total of 10 tasks represented as desired locations in the environment must be visited by four robots. The MRTA method that we used is the Hoplites-based coordination method explained in III-B. Each scenario has been tested 10 times in simulation and only MRTA metrics, i.e.,  $M_1$  and  $M_2$  have been considered. The goal of this study is to provide insight into the MRTA performance and behavior when humans exist in the environment. It is clear that without accounting for the human presence, social constraints have been frequently violated especially as the number of people increases. For the following two case studies, social metrics as well as the standard MRTA metrics have been considered.

2) *Comparative Evaluation in Simulation*: In this case study, a problem consisting of five tasks, four robots and three humans has been considered. The aim of this assessment is to study the performance of the socially-aware task planner in comparison to the other approaches. This case study demonstrates how team plans can change if social constraints are to be considered and how a robot that is stopped due to social interactions with people, can affect the team performance if a team collaboration strategy is not

foreseen for such circumstances. As depicted in Fig. 3, there exist a static human and two dynamic humans who start moving towards Robot 3, when the simulation begins. They initiate an interaction with the robot and therefore, the robot will be interrupted while moving to its assigned task.

To compare the performance of different MRTA strategies, four series of tests across two scenarios have been conducted. In *Scenario 2A* there are no humans present, whereas in *Scenario 2B* three humans exist in the environment as explained in the case study description. This latter scenario adopts three different algorithms: *Scenario 2B-AG*: human-agnostic robots, in this scenario humans are not considered in the team plan and are only considered as obstacles in robot navigation. *Scenario 2B-AWI*: human-aware robots as individuals, in this scenario humans are not considered in the team plan but navigation considers their presence as described in Section III-C. *Scenario 2B-AWT*: human-aware robots as a team, here humans are considered at both planning and individual navigation levels.

We note that three types of timeouts have been implemented for the following situations, (i) time-out due to social interactions, (ii) time-out due to no progression in the assigned task, and (iii) time-out due to large localization errors and lost robot. We have added the time-out for case (ii) for enabling the robots to finish their mission in case one robot takes too long to accomplish a task due to unforeseen reasons in scenarios other than *Scenario 2B-AWT*. The values considered for these time-outs are 10, 30, and 30 seconds, respectively. The larger values are chosen due to the impact of the time-out; for instance, abandoning a task without coordination with teammates is costly and the algorithm should be more conservative in triggering such action.

The social metrics are obtained excluding the robot stopped due to engaging in interactions with humans. The concept of layered social costmaps in ROS<sup>3</sup> has been used for implementing the social costs for human-aware navigation. The value of a social cost for a given position varies between 0 and 255. The radius of the social costmap is 1 m. Similar to [18],  $\sigma_x = 0.255 m$  and  $\sigma_y = 1.5 \times \sigma_x$ . Intrusion of the areas in front of a human is more severely penalized using this formulation. All four sets of tests have been repeated for ten simulation runs. Robots are relying on their self-localization for computing the local balance functions and the evaluation metrics ( $M_1 - M_5$ ) have been obtained from ground truth values provided by the simulation.

3) *Comparative Evaluation with Real Robots*: The same comparative study of Section V-C.2 in terms of methods has

<sup>3</sup>Robotic Operating System (ROS).

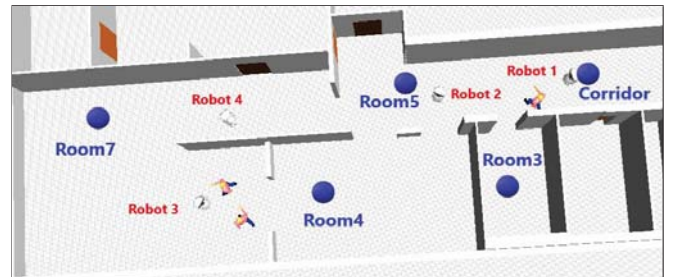


Fig. 3: Snapshot of a sample test for one static and two moving humans who are about to start interaction with Robot 3. There are five tasks depicted by blue circles representing the locations that should be visited by the robots.

<sup>2</sup>The FP7 European project MONarCH had its main robotic testbed at the Instituto Português de Oncologia de Francisco Gentil, Lisboa (IPOL).



Fig. 4: Snapshot of the real robot experiment.

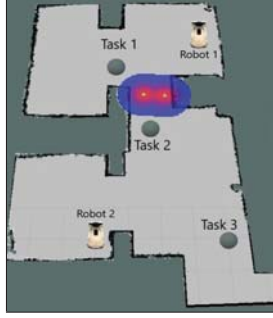


Fig. 5: Task, robots and human placement at the beginning of the experiment. Humans are facing each other. They are positioned in the center of the costmap. The areas closer to the humans and their interaction space are more costly (red) and the further away we get from the humans the cost will decrease (blue).

been conducted in reality. Here, a problem consisting of three tasks, two robots and two humans has been considered as depicted in Fig. 4. In this test, two static humans are having a conversation while standing in a rather narrow passage. The goal of this assessment is to examine how the social costs associated to humans and their interaction area can change the performance and plans of different MRTA approaches. The placement of tasks and the social costmaps are illustrated in Fig. 5. Robots are relying on their self-localization for computing the local balance functions. The position and orientation of the humans are given by a motion capture system consisting of 28 cameras with millimetric accuracy and is broadcasted to the robots. Again,  $\sigma_x = 0.255\text{ m}$  and  $\sigma_y = 1.5 \times \sigma_x$ . Experiments have been conducted in an arena of size  $11.5\text{ m} \times 7.0\text{ m}$ .

## VI. RESULTS AND DISCUSSION

In this section the results of the three case studies explained in the previous section will be discussed.

### A. Human-agnostic MRTA in Social Environments

Fig. 6 shows the performance of this test across different scenarios. The metrics used in this study are  $M_1$  and  $M_2$ , similar to [15]. It can be seen that both metrics are increased when humans are present. Since the environments are all exactly the same from the point of view of the task planning method, humans affect the timing ( $M_2$ ) of the tasks mostly as observed in Fig. 6. This is due to humans appearing as (dynamic) unmapped obstacles that could result in longer modified paths, blocked passages or interfere with the localization of the robots.

Based on Fig. 6b, the mission time for Scenario 1B and 1D with static people is higher compared to Scenario 1C and 1E with dynamic humans. This may seem counter-intuitive since

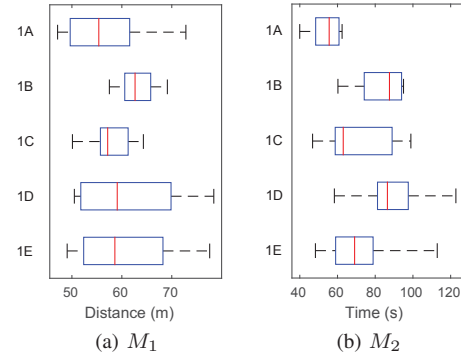


Fig. 6: The total distance and time for the five different scenarios for human-agnostic MRTA obtained from 10 runs.

human motion adds more complexity to the scenario. However, static humans cause a larger increase in the localization error of the robots. Indeed, throughout the runs we observed robots getting stuck in a close proximity of static humans. This happens since the robot thinks it is too close to an obstacle and stops moving, due to the localization error having jumps from time to time. Since static humans do not move, long periods of blockage that can only be broken by means of time-outs occur.  $M_2$  is increased in such cases since the task of the blocked robot is not reassigned to any other team member upon long waits. This is one reason that necessitates modification of team plans in social environments. In the case of moving humans, the robot is able to retrieve its correct position once it receives new information from the laser scans and when it senses a large enough distance to the surrounding obstacles. Thus, the increase in  $M_1$  and more so  $M_2$ , is less than the static case.

In our tests, increasing the number of humans (e.g., 1B vs. 1D and 1C vs. 1E) led to larger variations in  $M_1$ . The particular cases which lead to blockage, narrow passages and long waits, are the main cause of the larger traveled distances and longer mission times and do not necessarily increase as the number of people increases.

Since we positioned the static humans on challenging spots already with five humans (Scenario 1B), the overall difficulty of this scenario was similar to the case of ten static humans (Scenario 1D) in terms of problematic situations. As a result, the mean value of the two metrics are not very different as the number of humans are increased. This result has been certainly biased by the choice of human positions. However, this choice was made to ensure problematic situations exist in all scenarios. Nonetheless, in uncontrolled social environments more problematic situations are likely to happen as the number of humans increases.

### B. Comparative Evaluation in Simulation

To understand the task assignment depicted in Fig. 7, consider Fig. 3 again and assume Robot 1 has reached the Corridor task and Robot 2 is at Room 5. For deciding who is going to accomplish the next task, i.e., Room 3, without accounting for the human, the distance between Robot 1 at the Corridor is smaller than that of Robot 2 at Room 5. Therefore, in Scenario 3A and 3B-AG Robot 1 takes Room 3. In Scenario 3B-AWI, Robot 2 delegates Room 3 to Robot 1 through collaboration, despite becoming available earlier in time. In Scenario 3B-AWT on the other hand, Robot 2 takes Room 3 as the result of active coordination between Robot 1

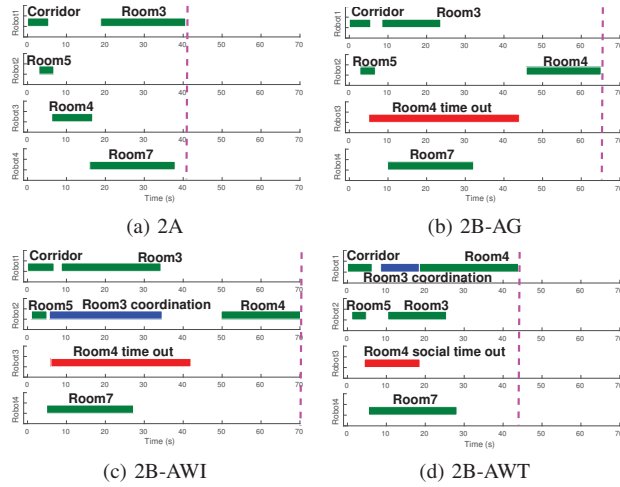


Fig. 7: Task assignment per robot over time for a sample run of the second set of experiments for scenarios 2A, 2B-AG, 2B-AWI, and 2B-AWT respectively. The blue blocks indicate collaboration requests and coordination and the red blocks represent long waits leading to time-out. End of mission ( $M_2$ ) is marked by the vertical line.

and Robot 2. Note that the team plan changes when social costs are taken into account.

For the second problematic situation leading to time-outs, in Scenario 2B-AG and 2B-AWI, Room 4 will be taken by the closest available robot which is Robot 2 but only after the time-out period. In Scenario 2B-AWT, the moment the social time-out for Robot 3 is issued, replanning through active coordination takes place and Robot 1 which is the closest available robot, will take the task (Robot 2 has been sent to Room 3). This significantly reduces the mission time ( $M_2$ ) compared to 2B-AWI as seen in Fig. 8b.

Fig. 8 demonstrates  $M_1$ - $M_5$  for Scenario 2A to 2B-AWT. For Scenario 2A, we expect to have the smallest  $M_1$  and  $M_2$  due to no humans existing in the environment. This is confirmed by Fig. 8a-b. In the human-agnostic method, robots do not modify their paths to respect the social spaces around humans and will therefore, travel a smaller distance compared to a human-aware robot for the same plan. Thus,  $M_1$  for Scenario 2B-AG will be smaller compared to 2B-AWI and 2B-AWT, as confirmed by Fig. 8a. However, the human-agnostic approach leads to large social costs due to not respecting social constraints. This can be observed in Fig. 8c-e where larger  $M_3 - M_4$  are recorded for the human-agnostic method. On the other hand, this method achieve the smallest values on  $M_5$  due to humans being considered merely as obstacles.

We can see that both Scenario 2B-AWI and 2B-AWT have negligible social costs and social zone violations. In principle, they should have zero  $M_3$  and  $M_4$ , however, small localization errors can lead to occasionally moving into areas with non-zero social costs or generating the false positive assumption of being there.  $M_5$  for both Scenario 2B-AWI and 2B-AWT is in the recommended range and larger compared to that of Scenario 2B-AG. We can see that on average, the robots navigate as close as the edge of the costmap (0.7 m) in Scenario 2B-AWI and 2B-AWT.

### C. Comparative Evaluation with Real Robots

Similar to the previous section, sample team plans for all four scenarios can be found in Fig.10. We can see how the team plan changes across different scenarios. Fig.10a displays the team plan in a human-free environment where

tasks are scored inversely proportional to their distance to the robot. In this experiment, the passage leading from Task 1 to Task 2 is populated by two interacting humans. Thus, large social costs are associated to that area. This causes Task 2 to be very costly for Robot 1 from the point of view of a social coordination mechanism. Additionally, this cost is large enough to stop the robot from moving and hence, causing a time-out. In Scenario 3B-AG and 3B-AWI, Task 2 is assigned to Robot 1 as it is closer to that location (see Fig. 5). However, since the passage can not be traversed, Robot 2 will take the task only after a progress time-out has been broadcasted. For Scenario 3B-AWI, we observed Robot 1 constantly moving to find a way to Task 2 or slightly moving in social zones and stopping there in some runs. This explains the larger  $M_1$  and  $M_3$  in Fig. 9.

In Scenario 3B-AWT, Robot 1 will initially take Task 2 since it becomes available before Robot 2. However, upon computing the value for the task a collaboration request is sent and Task 2 is delegated to Robot 2 due to its large social cost. Scenario 3B-AWT is shown to have superior performance in terms of  $M_1 - M_5$  compared to Scenario 3B-AWI while ensuring no social constraints are violated and maintaining a large enough  $M_5$  as shown in Fig. 8c-e. Not modifying robot trajectories to avoid social zones causes  $M_1$  and  $M_2$  to be smaller for Scenario 3A and 3B-AG at the cost of having large social costs. A video of sample runs of this test as well as further material related to this project can be found following the link indicated in the footnote<sup>4</sup>.

## VII. CONCLUSION AND FUTURE WORK

In this work, we have demonstrated the essential role of accounting for social factors in coordination mechanisms adopted by MRTA. Evaluating the performance of a human-agnostic MRTA approach in the presence of multiple static/moving people showed how the mission time can considerably increase if humans are present in the environment.

Simulation and real robot results confirm that accounting for social costs and time-outs that are the products of social encounters, on both distributed planning and individual navigation levels have superior results in social environments. This is done by extending our Hoplites-based coordination method by means of including social costs in the balance functions and bids, and requesting replanning and plan adjustment via active coordination for social time-outs.

Social environments can be very dynamic and unpredictable. Therefore, assuming that task costs can be correctly estimated for a given period of time is not always correct. For future steps of this research, effective bid estimation and rebidding mechanisms will be studied. Additionally, in real environments, global perception of the human poses are often not available and accessible to all robots. Examining the effects of partial information and local perception is another aspect that we will consider in the future steps of this work.

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<sup>4</sup><http://disal.epfl.ch/research/SocialRoboticsNavigation>

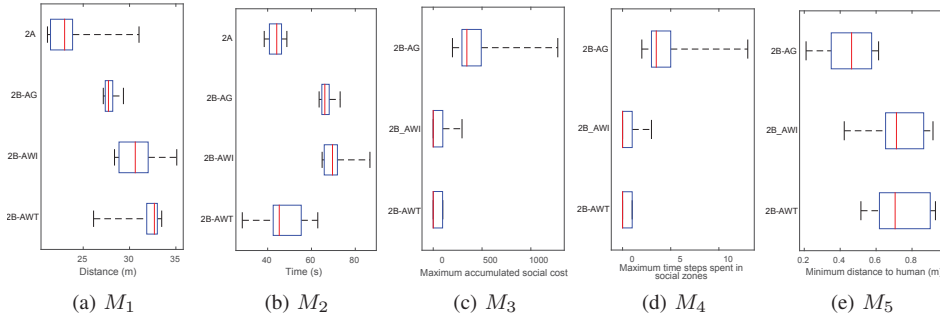


Fig. 8: Performance metrics for the second set of experiments in simulation obtained from 10 runs. Note that for metric  $M_3$  to  $M_5$ , no performance for scenario 2A is plotted since no human was present.

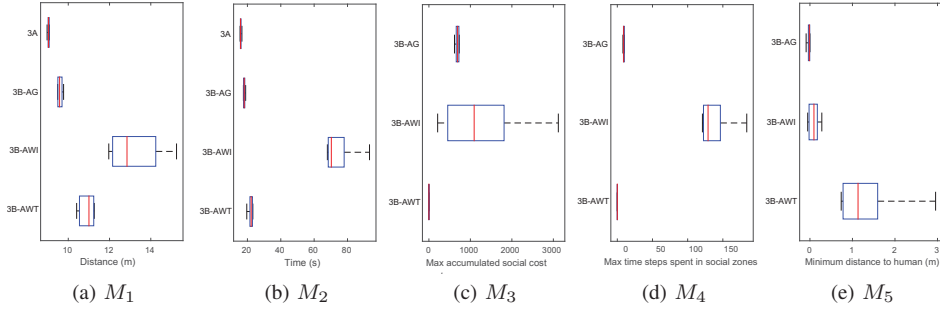


Fig. 9: Performance metrics for the third set of experiments for real robots obtained from five runs. Note that for metric  $M_3$  to  $M_5$ , no performance for scenario 3A is plotted since no human was present.

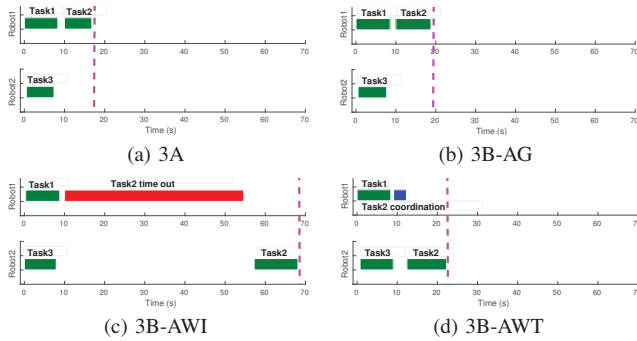


Fig. 10: Task assignment per robot over time for a sample run of the real robot experiment, for scenarios: 3A, 3B-AG, 3B-AWI, and 3B-AWT respectively. The blue block indicates collaboration requests and coordination and the red block represents long waits leading to time-out. End of mission ( $M_2$ ) is marked by the vertical line.

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